3.2. Navigating with Q-learning

In 2013, DeepMind published a paper entitled “Playing Atari with Deep Reinforcement Learning” that outlined their new approach to an old algorithm, which gave them enough performance to play six of seven Atari 2600 games at record levels. Crucially, the algorithm they used only relied on analyzing the raw pixel data from the games, just like a human would. This paper really set off the field of deep reinforcement learning.

The old algorithm they modified is called *Q-learning*, and it has been around for decades. Why did it take so long to make such significant progress? A large part is due to the general boost that artificial neural networks (deep learning) got a few years prior with the use of GPUs that allowed the training of much larger networks. But a significant amount is due to the specific novel features DeepMind implemented to address some of the other issues that reinforcement learning algorithms struggled with. We’ll be covering it all in this chapter.

3.2.1. What is Q-learning?

What is Q-learning, you ask? If you guessed it has something to do with the action-value function $Q_\pi(s,a)$ that we previously described, you are right, but that’s only a small part of the story. Q-learning is a particular method of learning optimal action values, but there are other methods. That is to say, value functions and action-value functions are general concepts in RL that appear in many places; Q-learning is a particular algorithm that uses those concepts.

Believe it or not, we sort of implemented a Q-learning algorithm in the last chapter when we built a neural network to optimize the ad placement problem. The main idea of Q-learning is that your algorithm predicts the value of a state-action pair, and then you compare this prediction to the observed accumulated rewards at some later time and update the parameters of your algorithm, so that next time it will make better predictions. That’s essentially what we did in the last chapter when our neural network predicted the expected reward (value) of each action given a state, observed